International Journal of Epidemiology, 2017, 1–3
doi: 10.1093/ije/dyx086
Letter to the Editor

Letter to the Editor

On wagging tales about causal inference

Ashley I. Naimi

Department of Epidemiology, University of Pittsburgh, Graduate School of Public Health, Parran Hall, 130 DeSoto Street, Pittsburgh, PA 15261. E-mail: ashley.naimi@pitt.edu

Krieger and Davey Smith (KDS) have written a lengthy polemic on the increasingly popular use of methods for causal modelling ('causal inference') in epidemiology.[1] This paper has instigated a lively discussion on the use of counterfactual modelling in epidemiology.[2–8] Yet much confusion remains. For example, in their last editorial for the *Journal*, Ebrahim *et al.*[9] state that 'the potential outcomes approach can be seen as … too far removed from many of the complex 'dirty' problems … of social epidemiology' (page 1704), but also note that '(c)ounterfactual methods … might help drive a new wave of social epidemiology focused on reducing social inequalities in health' (page 1708).

Confusion about the nature, scope, and purpose of counterfactual modelling is understandable.[10] It is a novel area of research with much progress to be made. As many have argued, confusion is also prominent feature of the critique by KDS. For example, the claim that the causal modelling framework requires that one view 'race' as an 'inherent feature of individuals'' (page 14) is simply not true. Daniel *et al.*[8] do an excellent job at pointing out the problems with this perspective, but they do not mention that this same point of confusion was raised by Krieger over a decade ago[11] and, despite numerous accounts that have since clarified the point,[12–18] it appears again in KDS.

One possible explanation for this continued confusion is the choice by KDS to emphasize parts of the quotes they display in their Table 4 over others. For example, KDS quote text from a recent review by *Glass et al.*,[19] in which the authors ask whether we should 'consider causal questions about inherent features of the individual … that cannot be reasonably translated into hypothetical interventions'. This latter part of the quote is, in fact, the core issue. But KDS fail to mention this core issue when they discuss race in the causal modelling framework. Causal modelling is agnostic about whether 'race' is an inherent feature of individuals, or a property of the social, economic, political and historical circumstances in which individuals live and die.[16–18] This point is well explained by Daniel *et al.*[8].

However, KDS take the argument further and accuse those who adopt the position that 'race' is not causal of committing 'bad biology and bad social science' (p 14). They support this by noting that 'epidemiological evidence provides supportive evidence for the hypothesis that race relations cause changes in the distributions of health' (p 14). This is the very same line of reasoning previously raised by Krieger,[11] and is another explanation for the continued confusion, as it is based on a well-known logical fallacy.

In the empirical literature, 'race' is routinely used to connote simple self-reported classification schemes based on the ill-conceived categories proposed by Blumenbach in the 19th century.[20] Considering the possible interpretations of the word 'race', KDS premise their argument on the informal fallacies of vagueness[21]/equivocation[21] when they switch from 'race' to 'race relations' in their critique. The problems with self-reported racial classification schemes are well documented,[14,22–25] and these problems are precisely why 'race' cannot be causal.

Race relations, on the other hand, can be affected by social policy and a host of other interventions.[26] Such interventions can be incorporated into the causal modelling framework without issue. Indeed, all the examples of evidence that KDS muster to support their claims, including the Civil Rights Act of the mid 1960s, the desegregation of hospitals, and the repeal of unjust laws are (policy) interventions, far removed from the simplistic classification schemes commonly used to represent 'race'. These examples provide support for the very claim that KDS oppose: 'race' is not causal. Social policies and interventions

© The Author 2017; all rights reserved. Published by Oxford University Press on behalf of the International Epidemiological Association

that affect social norms are, and researchers would benefit greatly from approaching their study using the formal approach to counterfactual modelling.

A second point of confusion not addressed by any of the responses to KDS is the claim that causal modelling is an approach 'driven solely by (deductive) logic or probabilities' (page 6). KDS contend that decisions to implement policies should be based on abductive logic. In several places, they describe abductive logic (defeasible reasoning, inference to the best explanation) as an 'alternative' (page 5) to the counterfactual approach. As noted by KDS, abduction proceeds as follows:

i. D is a collection of data/evidence/facts/observations;
ii. H explains D;
iii. No H′ explains D as well as H; therefore, H is probably true.

In fact, counterfactual modelling is closely patterned after the above. To illustrate, consider that data (D) are often collected to evaluate the magnitude of a given exposure-outcome relation, with the choice of the exposure and confounders being governed by a particular hypothesis (H) about the system under study. After estimating the exposure-outcome association, causal modelling requires careful consideration of alternative explanations (H′). These alternative explanations are often articulated as assumptions that include (but are not necessarily limited to) positivity, counterfactual consistency, no interference, no unmeasured confounding, no selection bias, correct model specification, and no measurement error. Because most of these assumptions cannot be verified, if they are assumed to hold, the responsible causal modeller would then state that 'H is probably true'.

A final issue is the nature of the critique levelled by KDS, as well as others.[9,27–29] In several instances, including their response to published comments,[29] KDS contend that the above-mentioned assumptions are too restrictive and do not correspond to the messy realities facing epidemiologists. Consequently, causal modelling is made out to be too disconnected from complex public health problems to yield meaningful insights.

There are several problems with this line of reasoning, the most important being that the order of the propositions in this logic is incorrect. Causal modelling does not begin with assumptions. Rather, given an effect of interest, available data and other conditions, causal modelling provides insight into the assumptions required to interpret statistical associations as counterfactual contrasts. Framed this way, the benefits of the approach become clearer, particularly in 'messy' public health settings. If, in a given instance, the assumptions required are seen as too stringent or far-fetched, the next step would be to judge whether the effect of interest should be defined differently, more or different data should be collected, alternative modelling approaches should be used or whether the results should be reported with a particular emphasis on the nature of the assumptions required.

Much of the opposition and concern KDS raise with regards to the causal modelling framework is based on misinformation and misrepresentation. Rather than suggest that epidemiologists pursue alternative routes, or forego causal modelling altogether, perhaps more attention should be given to the causal modelling framework, so that its principles and premises will be better understood.

Funding

This work was supported in part by NIH Grant Number UL1TR001857.

References

1. Krieger N, Davey Smith G. The tale wagged by the DAG: broadening the scope of causal inference and explanation for epidemiology. *Int J Epidemiol* 2016;**46:**1787–808.
2. VanderWeele TJ. Commentary: On causes, causal inference and potential outcomes. *Int J Epidemiol* 2016;**45:**1809.
3. Lawlor DA, Tilling K, Davey Smith G. Triangulation in aetiological epidemiology. *Int J Epidemiol* 2016;**45:**1866.
4. Blakely T, Lynch J, Bentley R. Commentary: DAGs and the restricted potential outcomes approach are tools, not theories of causation. *Int J Epidemiol* 2016;**45:**1835.
5. Weed DL. Commentary: Causal inference in epidemiology: potential outcomes, pluralism and peer review. *Int J Epidemiol* 2016;**45**:1838.
6. Robins JM, Weissman MB. Commentary: Counterfactual causation and streetlamps: what is to be done? *Int J Epidemiol* 2016;**45**:1830.
7. Broadbent A, Vandenbroucke JP, Pearce N. Response: Formalism or pluralism? A reply to commentaries on`Causality and causal inference in epidemiology'. *Int J Epidemiol* 2016;**45:**1841.
8. Daniel RM, De Stavola BL, Vansteelandt S. The formal approach to quantitative causal inference in epidemiology: misguided or misrepresented? *Int J Epidemiol* 2017, Jan 27. doi: 10.1093/ije/dyw227. [Epub ahead of print.]
9. Ebrahim S, Ferrie JE, Davey Smith G. The future of epidemiology: methods or matter? *Int J Epidemiol* 2016;**45:**1699.
10. Greenland S. For and against methodologies:some perspectives on recent causal and statistical inference debates. *Eur J Epidemiol* 2017;**32:**3–20.
11. Krieger N. Does racism harm health? Did child abuse exist before 1962? On explicit questions, critical science, and current controversies: an ecosocial perspective. *Am J Public Health* 2003;**93:**194–99.
12. Kaufman JS, Cooper RS. Commentary: Considerations for use of racial/ethnic classification in etiologic research. *Am J Epidemiol* 2001;**154:**291–98.
13. Kaufman JS. Epidemiologic analysis of racial/ethnic disparities: some fundamental issues and a cautionary example. *Soc Sci Med* 2008;**66:**1659–69.

14. Holland PW. Causation and race. In: Zuberi T, Bonilla-Silva E (eds). *White Logic, White Method: Racism and Methodology*. Plymouth, UK: Rowman & Littlefield, 2008.
15. Smith HL. Research design: toward a realistic role for causal analysis. In: Morgan SL (ed). *Handbook of Causal Analysis for Social Research*. New York, NY: Springer, 2013.
16. Naimi AI, Kaufman JS. Counterfactual theory in social epidemiology: reconciling analysis and action for the social determinants of health. *Curr Epidemiol Rep*. 2015;**2:**52–60.
17. Naimi AI. The counterfactual implications of fundamental cause theory. *Curr Epidemiol Rep* 2016;**3:**92–97.
18. Naimi AI, Schnitzer ME, Moodie EEM *et al*. Mediation analysis for health disparities research. *Am J Epidemiol* 2016;**184:** 315–24.
19. Glass TA, Goodman SN, Hernan MA *et al*. Causal inference in public health. *Annu Rev Public Health* 2013;**34:**61–75.
20. Jacobson MF. *Whiteness of a Different Color: European Immigrants and the Alchemy of Race*. Cambridge, MA: Harvard University Press, 1998.
21. Hurley PJ. *A Concise Introduction to Logic*. Belmont, CA: Wadsworth, 2011.
22. Zuberi T. Deracializing social statistics: problems in the quantification of race. *Ann Am Acad Polit Soc Sci* 2000;**568:**172–85.
23. James A. Making sense of race and racial classification. In: Zuberi T, Bonilla-Silva E (eds). *White Logic, White Method: Racism and Methodology*. Plymouth, UK: Rowman & Littlefield, 2008.
24. Marshall GA. Racial classifications: popular and scientific. In: Harding S (ed). *The 'Racial' Economy of Science :Toward a Democratic Future*. Bloomington, IN: Indiana University Press, 1993.
25. LaVeist TA. Beyond dummy variables and sample selection: what health services researchers ought to know about race as a variable. *Health Serv Res* 1994;**29:**1–16.
26. Feagin JR. *Systemic Racism: A Theory of Oppression*. New York, NY: Routledge, 2006.
27. Vandenbroucke JP, Broadbent A, Pearce N. Causality and causal inference in epidemiology: the need for a pluralistic approach. *Int J Epidemiol* 2016;**45:**1776.
28. El-Sayed AM. Complex systems for a complex issue: race in health research. *Virtual Mentor* 2014;**16:**450–54.
29. Krieger N, Davey Smith G. Response: FACEing reality: productive tensions between our epidemiological questions, methods and mission. *Int J Epidemiol* 2016;**45**:1852.